

Coding Portfolio

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Five projects drawn from research and independent work, spanning data cleaning and visualization, survey analysis, causal inference, and statistical simulation. Each section gives brief context, followed by representative code and the output it produced; complete do-files are linked below.

<https://github.com/sak194-sketch/Coding-Portfolio->

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1. Minimum Wage Trends in Bangladesh

World Bank · Development Research Group

Built from wage schedules published across roughly fifty industries on Bangladeshi government websites, this do-file cleans the raw records into a Stata-compatible panel and charts industry wage trends over time. The output supported cross-country minimum-wage research within the World Bank's Development Research Group. Because so many industries are tracked at once, the graphing routine batches them six at a time so the variation in wages stays legible on each chart.

```
* Clean monthly wage and dates
destring monthly_rate, replace ignore(", BDT") force
gen date2 = date(date_start, "DMY")
format date2 %td
drop date_start
rename date2 date_start

* Workers only, grouped six industries per graph
keep if lower(trim(worker_category)) == "worker"
collapse (mean) monthly_rate, by(industry date_start _ext)
egen ind_id = group(industry)
local n_groups = int(ceil(`n_inds' / 6))

forvalues g = 1/`n_groups' {
    * ... build (line) + (scatter) layers for each industry in the group ...
    twoway `plots', legend(order(`leg') size(small) cols(1)) ///
        xtitle("Year") ytitle("Monthly Minimum Wage (BDT)") ///
        title("Minimum Wages by Sector (Group `g' of `n_groups')") ///
        subtitle("Bangladesh - Workers")
    graph export "$output/bgd_workers_group`g'.png", replace
}
```

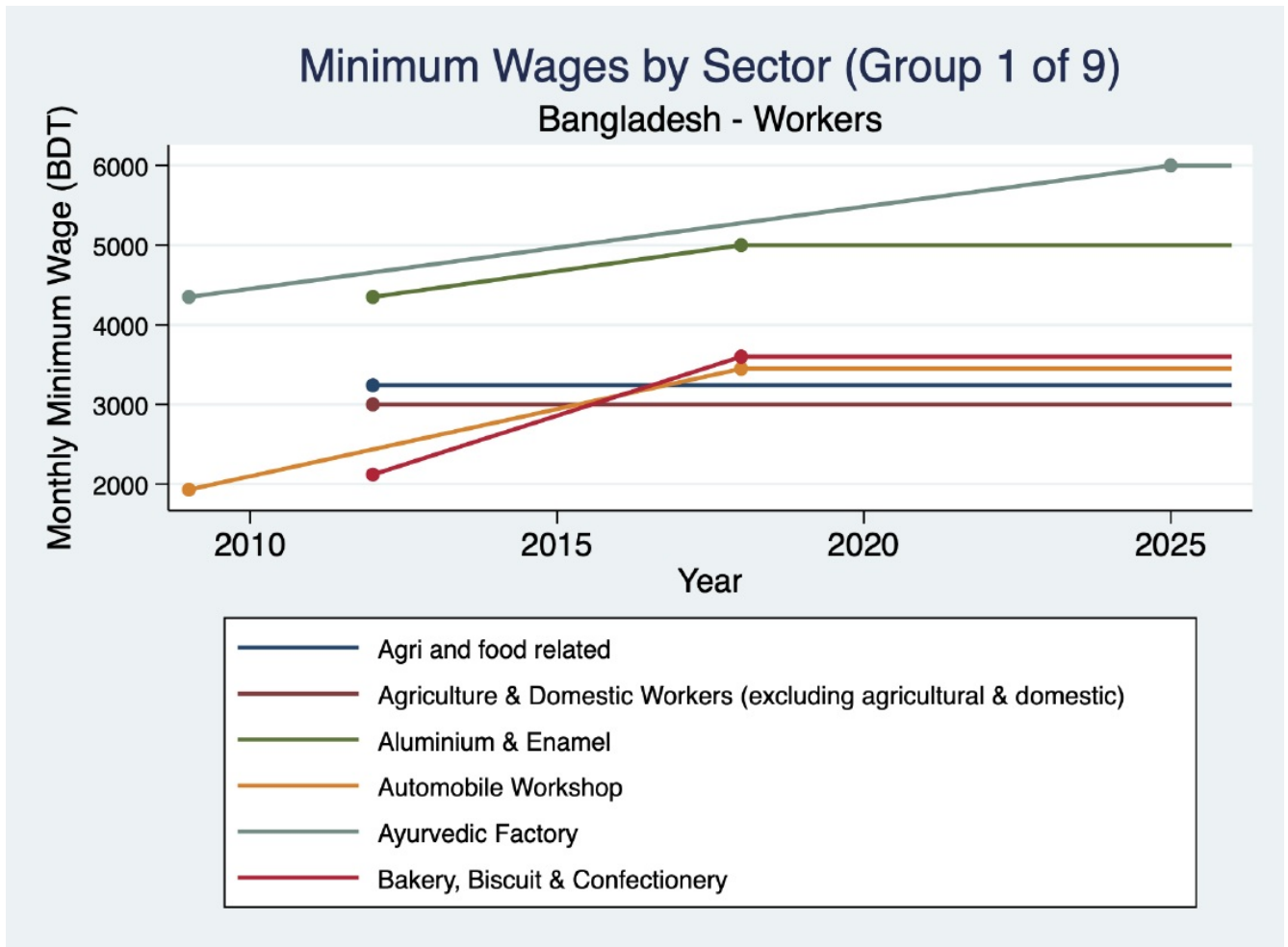


Figure 1. Output of the code above: the first of nine grouped charts, showing six industries' minimum-wage trajectories through 2025.

The garment sector (not shown in this group) has the most variation of any industry: each step change lines up with a national event: a 2013 wage increase following the Rana Plaza collapse, a 2020 dip during COVID-19, and a 2023 increase following national wage protests.

2. Political Participation & Education Attitudes in Punjab

LEAPS · Center for Economic Research in Pakistan

A baseline-survey analysis of non-electoral political participation across two districts of Punjab, examining how support programs, perceptions of education quality, civic engagement, and socioeconomic status relate to one another prior to any intervention exposure. The excerpt below constructs a household wealth index from asset ownership and visualizes its distribution across the sample.

```
* HH wealth index: assign 'poor'/'rich' by asset ownership, then run
* a PCA / factor model and split households into quintiles
predict asset_wealth_index
xtile asset_wealth_quintile = asset_wealth_index, nq(5)
label define wealth 1 "Poorest" 2 "Poorer" 3 "Middle" 4 "Richer" 5 "Richest"
label values asset_wealth_quintile wealth

* Distribution of the index
histogram asset_wealth_index, bin(30) frequency ///
    title("Distribution of Household Wealth Index") ///
    xtitle("Wealth Index") ytitle("Number of Households") ///
    lcolor(navy) fcolor(navy%60)
graph export "wealth_histogram.pdf", replace
```

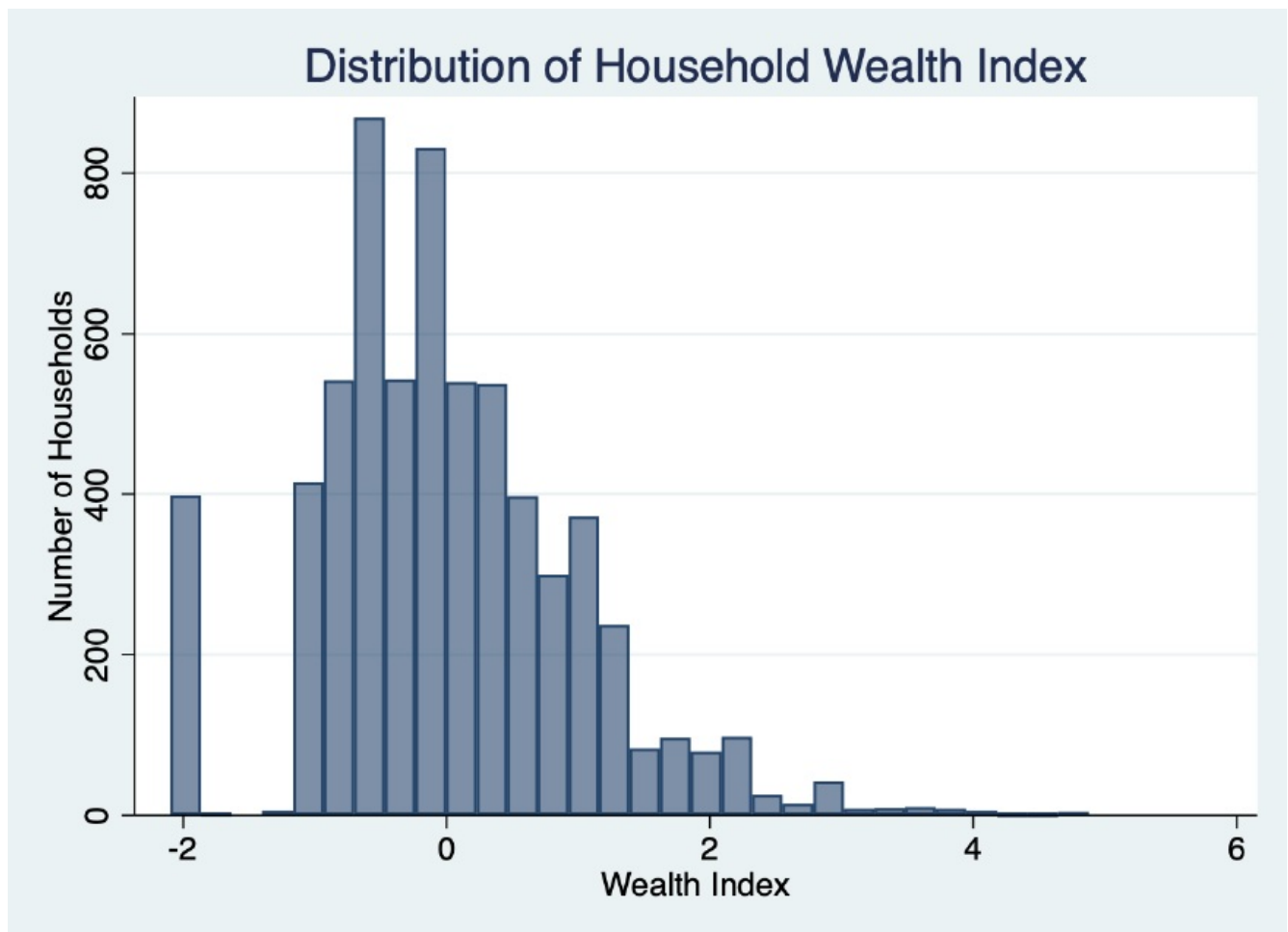


Figure 2. Output of the code above: the wealth index is right-skewed, with a small cluster of relatively poorer households near -2 .

The regression below tests whether the order in which “health” and “education” were presented as answer options predicted whether respondents selected them as a top-three issue, a check for response-order bias

in the survey instrument.

```
gen edu_health_top_three = 1 if s2_2q5a1==1 | s2_2q5a2==1 | s2_2q5a3==1 | ///
    s2_2q5b1==6 | s2_2q5b2==6 | s2_2q5b3==6
replace edu_health_top_three = 0 if edu_health_top_three == .

gen treat_health_edu_early = .
replace treat_health_edu_early = 1 if !missing(s2_2q5a1)
replace treat_health_edu_early = 0 if !missing(s2_2q5b1)

reg edu_health_top_three treat_health_edu_early
esttab using "regression_table.tex", replace se label ///
    title("Impact of Option Order on Choosing Health/Education")
```

3. Statistical Power & Hypothesis Testing Simulations

Independent work

A set of Monte Carlo simulations built to develop intuition for core ideas in statistical inference: Type I / Type II error rates, statistical power, confidence intervals constructed by inverting a hypothesis test, and how the prior probability that an effect is real interacts with significance testing.

Power, Type I and Type II error via simulation

```
cap prog drop balloon_power
prog def balloon_power, rclass
    args n pA pB
    clear
    set obs 1
    gen worldA = rbinomial(`n', `pA')
    gen worldB = rbinomial(`n', `pB')
    return scalar reds_wA = worldA
    return scalar reds_wB = worldB
end

postfile handle int n int crit_high float type1 type2 power using `results'
foreach n in 5 10 20 2000 {
    simulate grpA=r(reds_wA) grpB=r(reds_wB), reps(10000) nodots: ///
        balloon_power `n' 0.10 0.20
    * ... find critical value, then compute Type I / Type II error rates ...
    post handle (`n') (`crit') (`type1') (`type2') (`=1-`type2'')
}
```

Base-rate fallacy: prior probability vs. significance

```
local priors 0.05 0.10 0.25 0.50 0.95
foreach prior of local priors {
    simulate true=r(true) significant=r(significant), reps(10000) nodots: ///
        p221 0.05 `prior'
    qui tab true significant, matcell(m)
    scalar P_signif_given_true = m[2,2] / (m[2,1] + m[2,2])
    scalar P_nottrue_given_signif = m[1,2] / (m[1,2] + m[2,2])
    di "P(significant | true) = " %6.3f P_signif_given_true
    di "P(not true | significant) = " %6.3f P_nottrue_given_signif
}
```

At low priors (e.g. 5% of tested effects are real), even a well-powered test yields a majority of “significant” results that are false positives: a simulated illustration of the base-rate fallacy.

4. Long-Run Effects of Corporal Punishment: Attrition Reweighting, Peru

Center for Global Development · Georgetown University Capstone

Part of a four-country longitudinal study (Ethiopia, India, Peru, Vietnam) on the long-run effects of corporal punishment on later-life outcomes, co-authored with three classmates. My portion of the analysis covers Peru end-to-end: attrition testing, inverse-probability reweighting, and later the OLS/IV/heterogeneity/LASSO results in Project 5. The figure below is a team output shown for study context; it was not produced by the code excerpted in this portfolio.

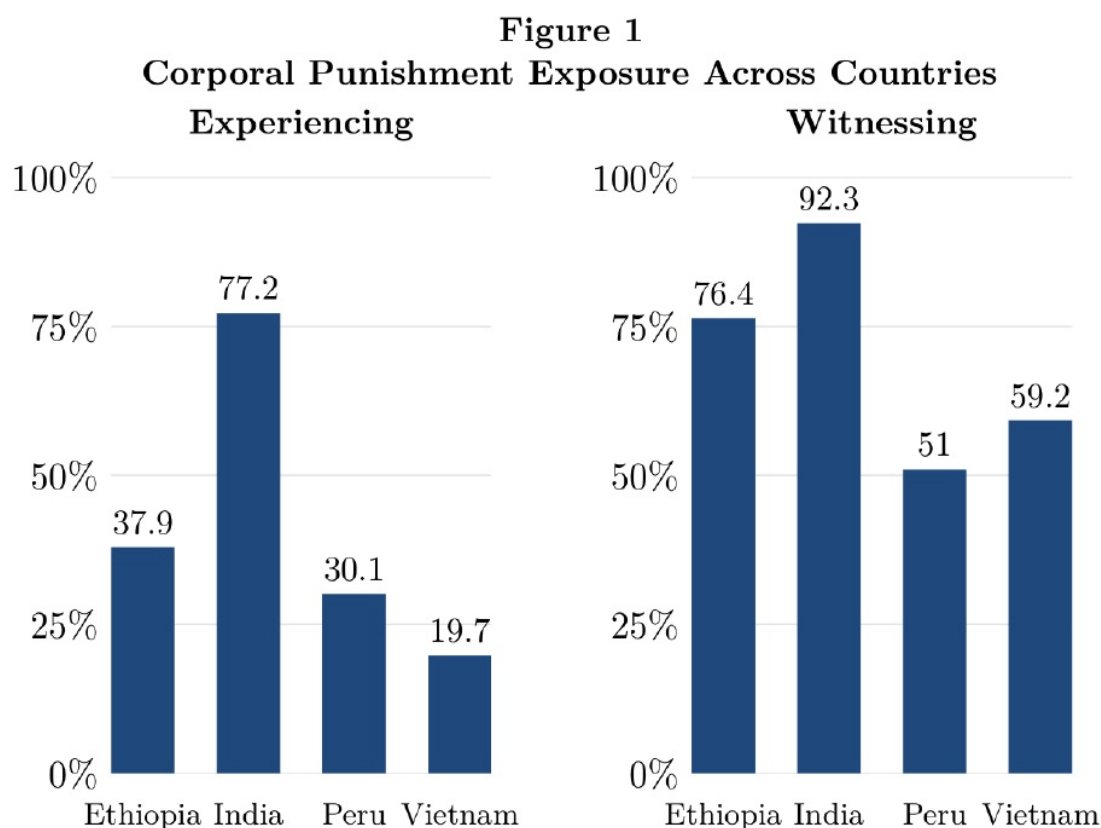


Figure 3. Team output, included for study context only, not produced by the code below. Source: Corporal Punishment and Later Life Outcomes: Evidence from the Young Lives Study (Duangphummet, Kashif, Roy & Sijapati, Georgetown University Capstone, 2026).

Attrition balance test, Peru

For each baseline covariate, continuous variables are min-max normalized to [0,1] so the mean difference between attriters and non-attriters is interpretable on a common scale, then compared using `iebal` tab.

```

* Min-max normalize continuous variables to [0,1]
foreach var in math ppvt agency_index_R3 shame_index_R3 tcphyur3 {
    quietly summarize `var'
    gen `var'_norm = (`var' - r(min)) / (r(max) - r(min))
}

label define attrite_lbl 0 "Not attrite" 1 "Attrite"
label values attrite_R5 attrite_lbl
label values attrite_R6 attrite_lbl

local baseline_outcomes math_norm ppvt_norm agency_index_R3_norm ///
    shame_index_R3_norm male tcphyur3_norm

iebalstab `baseline_outcomes', grpvar(attrite_R5) ///
    savetex("$tables/Attrition_pe_R5.tex") replace rowvarlabels
iebalstab `baseline_outcomes', grpvar(attrite_R6) ///
    savetex("$tables/Attrition_pe_R6.tex") replace rowvarlabels

```

Variable	Group 1 (Not Attrite)		Group 2 (Attrite)		(1)-(2)	
	N	Mean (SE)	N	Mean (SE)	N	Mean Diff.
Panel A. Round 5						
Math Score (Std.)	1770	0.492 (0.005)	93	0.489 (0.021)	1863	0.003
Vocabulary Score (Std.)	1724	0.469 (0.003)	93	0.470 (0.015)	1817	0.001
Agency Index	1802	0.579 (0.003)	95	0.585 (0.011)	1897	0.006
Shame Index	1802	0.288 (0.003)	95	0.293 (0.014)	1897	0.004
Male	1803	0.503 (0.012)	95	0.505 (0.052)	1898	0.002
Corporal Punishment	1803	0.167 (0.006)	95	0.189 (0.031)	1898	-0.023
Panel B. Round 6						
Math Score (Std.)	1692	0.499 (0.005)	171	0.423 (0.015)	1863	0.077***
Vocabulary Score (Std.)	1650	0.472 (0.003)	167	0.432 (0.012)	1817	0.040***
Agency Index	1724	0.582 (0.003)	173	0.562 (0.008)	1897	0.020**
Shame Index	1724	0.287 (0.003)	173	0.300 (0.010)	1897	-0.013
Male	1725	0.497 (0.012)	173	0.561 (0.038)	1898	-0.063
Corporal Punishment	1725	0.166 (0.006)	173	0.182 (0.022)	1898	-0.016

Table 1. Output of the code above. Peru shows the lowest attrition of the four countries in the study and, unlike Ethiopia, India, and Vietnam, shows no significant baseline imbalance at Round 5. By Round 6, attriters had significantly lower math, vocabulary, and agency scores, motivating the reweighting step below.

Inverse probability weighting

```
* Leave-one-out community average of CP exposure (used later as the IV)
bysort clustid: egen total_cp = total(tcphothr3)
bysort clustid: gen n_comm = _N
gen avg_cp_others = (total_cp - tcphothr3) / (n_comm - 1)

* IPW for Round 5: P(observed) = 1 - P(attrite), weight = 1 / P(observed)
logit attrite_R5 math_r3 ppvt_r3 agency_index_R3 male tcphyur3
predict p_obs_R5, pr
replace p_obs_R5 = 1 - p_obs_R5
gen ipw_R5 = 1 / p_obs_R5

* Trim at the 1st / 99th percentile to prevent extreme weights
egen p1_R5 = pctlile(ipw_R5), p(1)
egen p99_R5 = pctlile(ipw_R5), p(99)
gen ipw_R5_trim = ipw_R5
replace ipw_R5_trim = p1_R5 if ipw_R5 < p1_R5 & !missing(ipw_R5)
replace ipw_R5_trim = p99_R5 if ipw_R5 > p99_R5 & !missing(ipw_R5)
```

5. Long-Run Effects of Corporal Punishment: IV, Heterogeneity & LASSO, Peru

Center for Global Development · Georgetown University Capstone

Building on the reweighted Peru panel from Project 4, this stage estimates the causal effect of corporal punishment on cognitive, psychosocial, and later-life outcomes using an instrumental-variables strategy, tests whether the effect varies by subgroup, and uses LASSO to select baseline controls in a principled, data-driven way.

Heterogeneity analysis

```
gen tcp_male          = tcphyur3 * male
gen tcp_urban         = tcphyur3 * urban
gen tcp_caredu_juniorplus = tcphyur3 * careed_juniorplus

quietly areg cog_r5 tcphyur3 tcp_male $controls_het_male, ///
      absorb(clustid) vce(cluster clustid)
estimates store cog_male
* ... repeat for tcp_urban, tcp_caredu_juniorplus ...

esttab cog_male cog_urban cog_careed using "$stables/HET_cognitive_r5.tex", ///
      replace b(3) se(3) label star(* 0.10 ** 0.05 *** 0.01) booktabs ///
      title("Heterogeneity Analysis: Cognitive Outcomes (R5), Peru") ///
      mtitles("Gender" "Urban" "Caregiver Education") ///
      keep(tcphyur3 tcp_male tcp_urban tcp_caredu_juniorplus)
```

	(1) Male	(2) Urban	(3) Caregiver Ed. Above Primary
Corporal Punishment	-0.162*** (0.035)	0.000 (0.051)	-0.149*** (0.043)
CP x Male	0.107*** (0.037)		
CP x Urban		-0.167** (0.074)	
CP x Caregiver Ed.: Above Primary			0.065 (0.075)
Observations	1,803	1,803	1,803
R-squared	0.51	0.52	0.52

Table 2. Output of the code above. The cognitive penalty from corporal punishment is significantly smaller for male students (a positive interaction offsetting a negative baseline effect) and significantly larger for urban children.

Instrumental variables

```
* IV baseline and with cluster FE, plus first-stage F-statistic
quietly ivregress 2sls `outcome' (tcphyur3 = avg_cp_others) ///
    $controls_OLSwthclusterFE [pweight = ipw_R5_trim], vce(cluster clustid)
quietly estat firststage
matrix f = r(singleresults)
estadd scalar cd_f = f[1,4]
estimates store i2_`outcome'

esttab o1_`outcome' o2_`outcome' i1_`outcome' i2_`outcome' ///
    using "$tables/PE_combined_R5.tex", append ///
    mgroups("OLS" "IV", pattern(1 0 1 0) ///
        prefix(\multicolumn{@span}{c}{}) suffix{}) span) ///
    stats(N r2 cd_f cluster_fe, labels("Obs." "R2" "First-stage F" "Cluster FE"))
```

LASSO-selected controls

```
local total_union_R5 ""
foreach outcome of global outcomes_R5 {
    dsregress `outcome' tcphyur3, controls($full_controls) ///
        rseed(12345) vce(cluster clustid)
    local sel = e(controls_sel)
    local total_union_R5 : list total_union_R5 | sel

    quietly regress `outcome' tcphyur3 `sel', vce(cluster clustid)
    di "`outcome': N=" e(N) " | R2=" %4.2f e(r2) " | controls: `sel'"
}
di "UNION OF ALL LASSO-SELECTED CONTROLS (R5): `total_union_R5'"
```

Effect of corporal punishment, Round 5 (OLS vs. OLS + cluster FE)

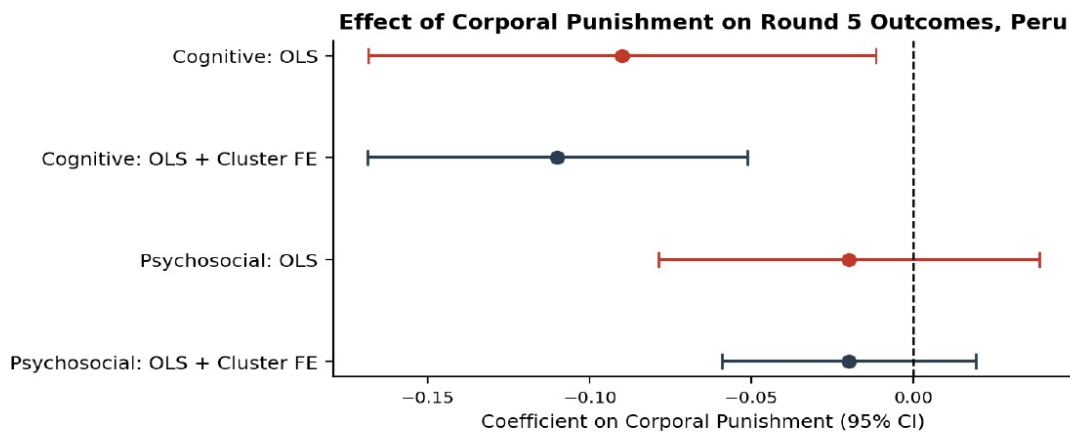


Figure 4. Coefficient plot built from the OLS and OLS+cluster-FE results for cognitive performance and the composite psychosocial (Anderson) index, Peru, Round 5. Cognitive performance is negative and significant under both specifications; the psychosocial effect is not distinguishable from zero.